

Faculty Leave Automation

by

Mutta Adithya Sai Srinivas 423158

Nannam Bhuvan 423161

Under the guidance of
Dr. KARTHICK S



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY, ANDHRA PRADESH
TADEPALLIGUDEM - 534102, INDIA
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Faculty Leave Automation

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*of
Bachelor of Technology*

by

Mutta Adithya Sai Srinivas 423158

Nannam Bhuvan 423161

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Mutta Adithya Sai Srinivas 423158

Nannam Bhuvan 423161

CERTIFICATE

It is certified that the work contained in this report titled “**FACULTY LEAVE AUTOMATION**” by “Mutta Adithya Sai Srinivas, bearing Roll No: 423158” and “Nannam Bhuvan, bearing Roll No: 423161” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Dr. KARTHICK S
Computer Science and Engineering
NIT Andhra Pradesh

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Mutta Adithya Sai Srinivas 423158

Nannam Bhuvan 423161

ABSTRACT

In our institution, the leave approval process involves multiple layers of approvals, which is time-consuming and inefficient if done manually. This abstract introduces the "Faculty Leave Automation", a dedicated digital solution developed to streamline the leave applications and approval workflow within the institution. This solution automates the existing manual process by ensuring transparency, accountability, and real-time leave tracking for all faculty.

Key features of this project include a centralized portal for submitting and tracking leave applications in real-time. It includes different leave categories abides with institutional rules and guidelines and utilizes role-based access control for faculty, heads of departments, deans, and administrative officials like the Registrar and Director. This project will route leave requests to the corresponding approvers based on predefined approval hierarchy. Security is ensured through institute email and OTP verification before direct access. This platform also features the real-time notifications, priority handling for urgent leave requests, leave balance tracking, and a dashboard for users.

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CHAPTER 1

1.1 Introduction

"Faculty Leave Automation" is a digital solution designed to streamline leave applications and approvals process at NIT Andhra Pradesh. The existing manual leave approval process often causes delays for faculty, deans, and heads. This solution automates the entire process, ensuring transparency and accountability, and allowing real-time leave tracking for all faculty. It aligns with NIT Andhra Pradesh's leave rules and guidelines, handles different types of leave and approval levels, simplifies the overall process, and reduces administrative workload.

1.2 Purpose

This project analyzes the "Faculty Leave Automation" at NIT Andhra Pradesh. Our aim is to know how it enhances the institution's administration and improves employee experience and efficiency in leave management. We're checking out what it can do, how it makes things automatic, how secure it is, and who uses it. This project's goal is to show how it solves problems with managing leaves. Ultimately, this project will inform the institution about the benefits of using this technology to handle tasks automatically.

CHAPTER 2

2.1 Problem Statement

The current leave approval process at NIT Andhra Pradesh is manual and paper-based, which leads to time consumption and inefficiency. Faculty members struggle with an unclear multi-level approval hierarchy, causing further frustration and confusion. Additionally, there is no mechanism for faculty to prioritize urgent leave requests, and no proper way to track leaves in real-time.

2.2 Requirements and Specifications

2.2.1 Software Requirements

- **Programming Languages:** TypeScript
- **Frontend Technologies:** HTML, Tailwind CSS, React.js with Vite.js
- **Backend Technologies:** Node.js with Express.js
- **Database:** PostgreSQL (Relational Database)
- **Development Tools:** Visual Studio Code, Postman

2.2.2 Technical Skills

- Proficiency in TypeScript is required for writing type-safe code.
- HTML proficiency is required for structuring web page content
- Mastery of Tailwind CSS is needed for utility-first, responsive UI design
- Skill in React.js with Vite.js is necessary for fast, modular frontend development
- Knowledge of Node.js and Express.js is needed for backend API creation and server-side logic
- Understanding of PostgreSQL is required for designing, managing, and querying relational databases

CHAPTER 3

3.1 Technical Overview

The front-end of this project was built using React.js to create a dynamic and responsive user interface, and Vite.js for fast loading and hot module replacement. TypeScript was used to ensure type safety. For the back-end, Node.js was used for managing logic, routing, and API handling, along with Express.js for server-side communication, and a PostgreSQL database. Upon completion, the project was planned for deployment on either Fly.io or Render, chosen for their simplicity and scalability. Additionally, Nodemailer was used to send real-time email notifications for OTP verification [7].

3.2 Database Design

For our project, we are using a PostgreSQL database, a relational database schema designed to efficiently manage and organize data [6] related to users, leave applications, approvals, notifications. The key tables and their relationships are outlined below in Figure 3.1.

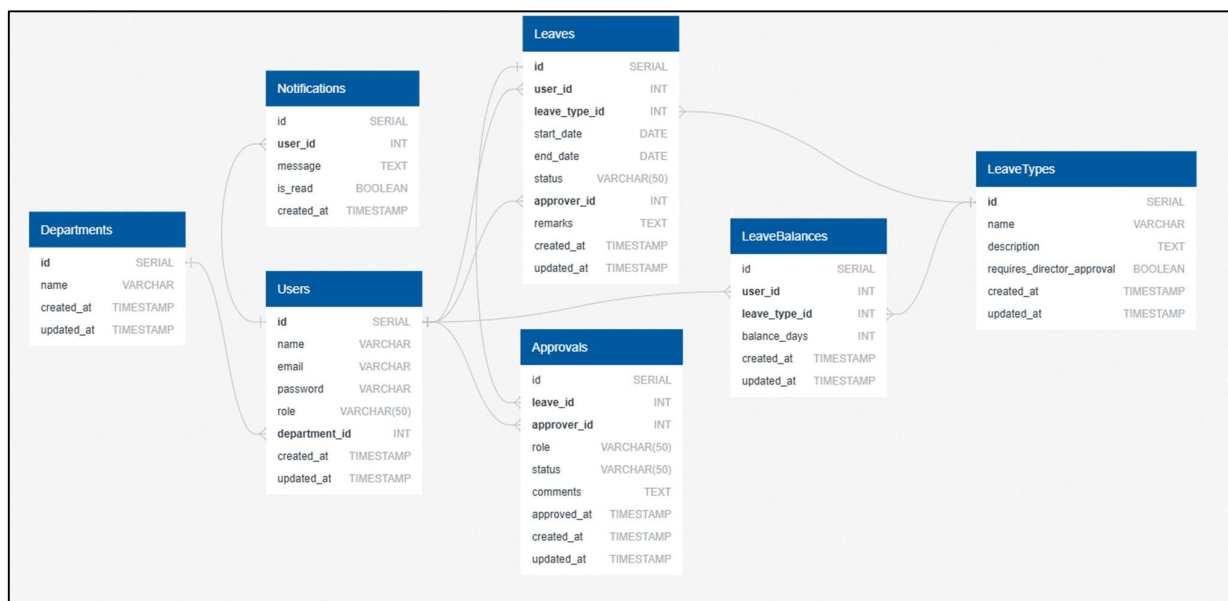


Figure 3.1 Database Design

The above database is in Boyce-Codd Normal Form (BCNF) which was highly optimized.

CHAPTER 4

4.1 User Authentication

The system ensures secure access through institute email verification and OTP verification. Users log in using their official institute email addresses, and a one-time password (OTP) is sent to their registered email for secure login verification. Role-based authentication assigns access levels based on the user's role (e.g., faculty, head, dean, registrar).

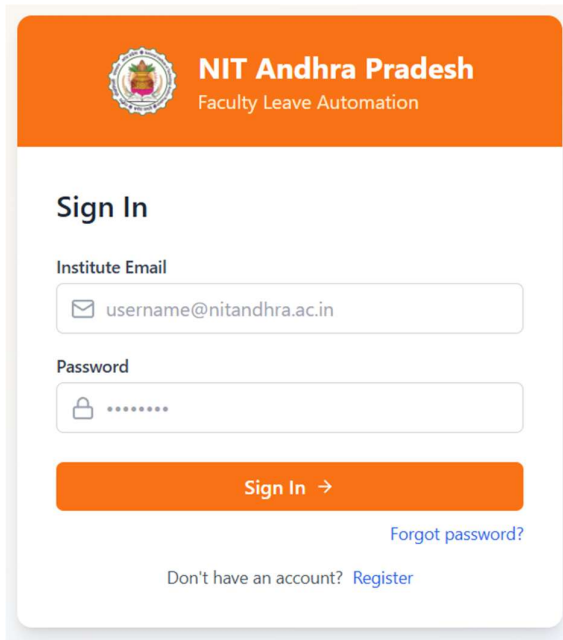
The image shows the 'Sign In' page of the NIT Andhra Pradesh Faculty Leave Automation system. At the top, there is an orange header with the NIT Andhra Pradesh logo and the text 'NIT Andhra Pradesh Faculty Leave Automation'. Below the header, the title 'Sign In' is displayed. There are two input fields: 'Institute Email' with a placeholder 'username@nitandhra.ac.in' and 'Password' with a placeholder '.....'. Below these fields is an orange button labeled 'Sign In →'. To the right of the button is a link 'Forgot password?'. At the bottom, there is a link 'Don't have an account? Register'.

Figure 4.1.1 Login Page

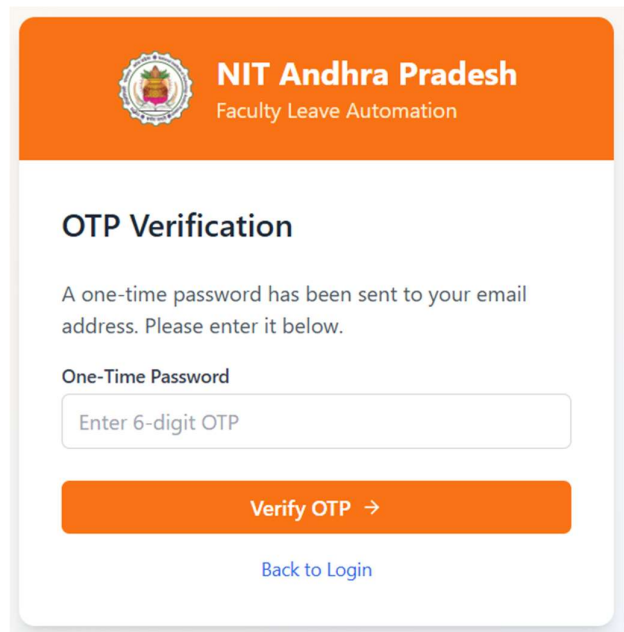
The image shows the 'OTP Verification' page of the NIT Andhra Pradesh Faculty Leave Automation system. At the top, there is an orange header with the NIT Andhra Pradesh logo and the text 'NIT Andhra Pradesh Faculty Leave Automation'. Below the header, the title 'OTP Verification' is displayed. A message states: 'A one-time password has been sent to your email address. Please enter it below.' Below this message is an input field labeled 'One-Time Password' with a placeholder 'Enter 6-digit OTP'. Below the input field is an orange button labeled 'Verify OTP →'. At the bottom, there is a link 'Back to Login'.

Figure 4.1.2 OTP Verification

4.2 Faculty side Features

4.2.1 Leave Application Submission

Faculty can submit leave applications using customized forms tailored to each leave type. The system includes pre-filled forms based on employee category and leave type, including essential fields as mandated by the rules (e.g., purpose of leave, supporting documents) as detailed process shown in Figure 4.2.1.1, Figure 4.2.1.2, and Figure 4.2.1.3.

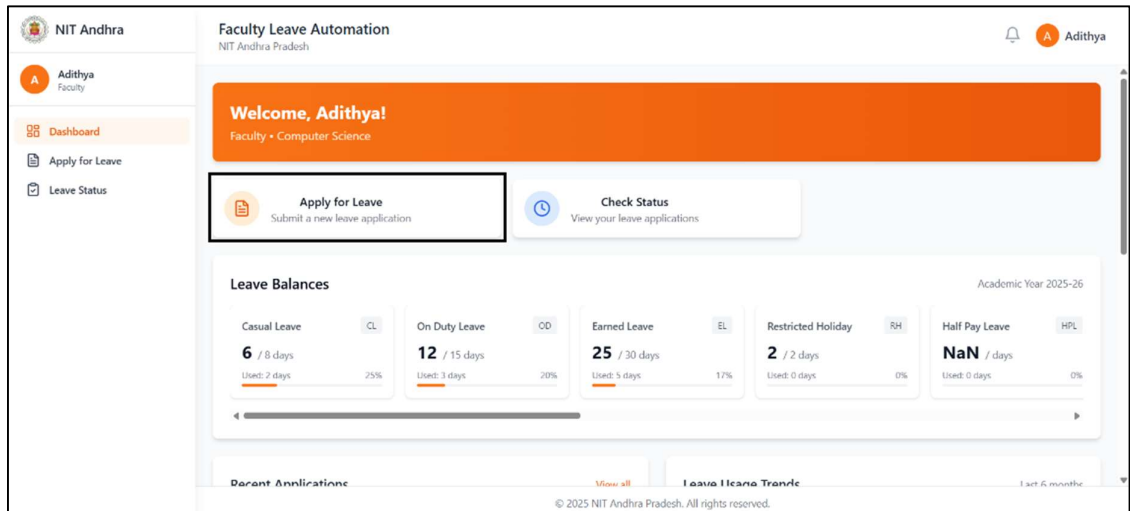


Figure 4.2.1.1 Apply for Leave Navigation

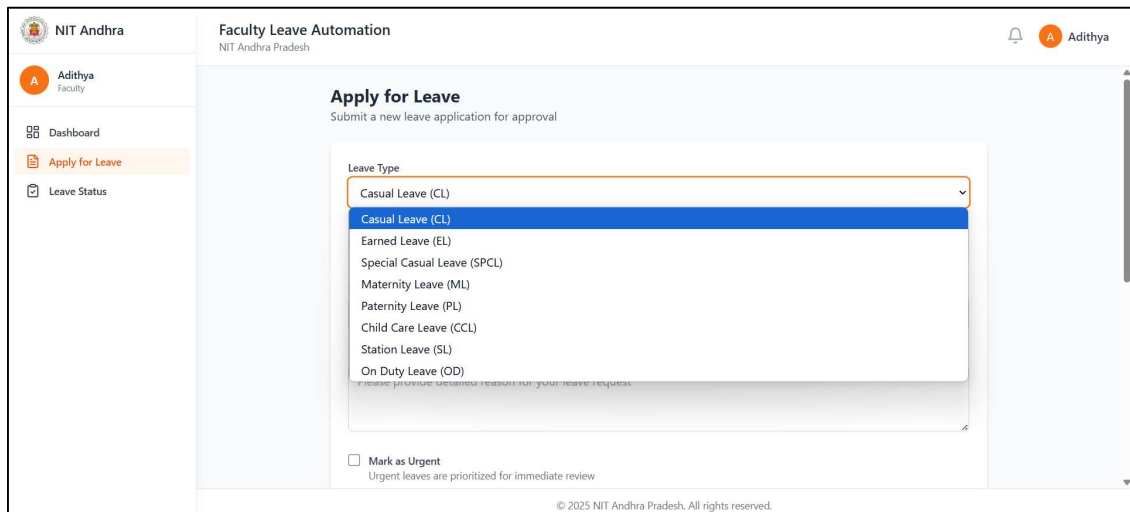


Figure 4.2.1.2 Apply for Leave Page (start)

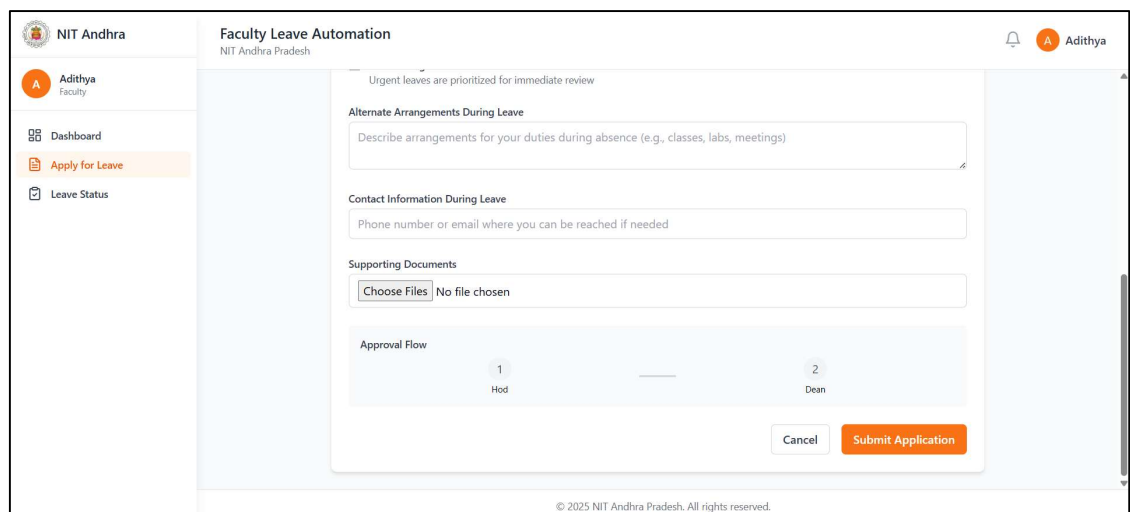


Figure 4.2.1.3 Apply for Leave Page (end)

4.2.2 Leave Balance and History

Faculty can view their remaining leave balances as shown in Figure 4.2.2.1.

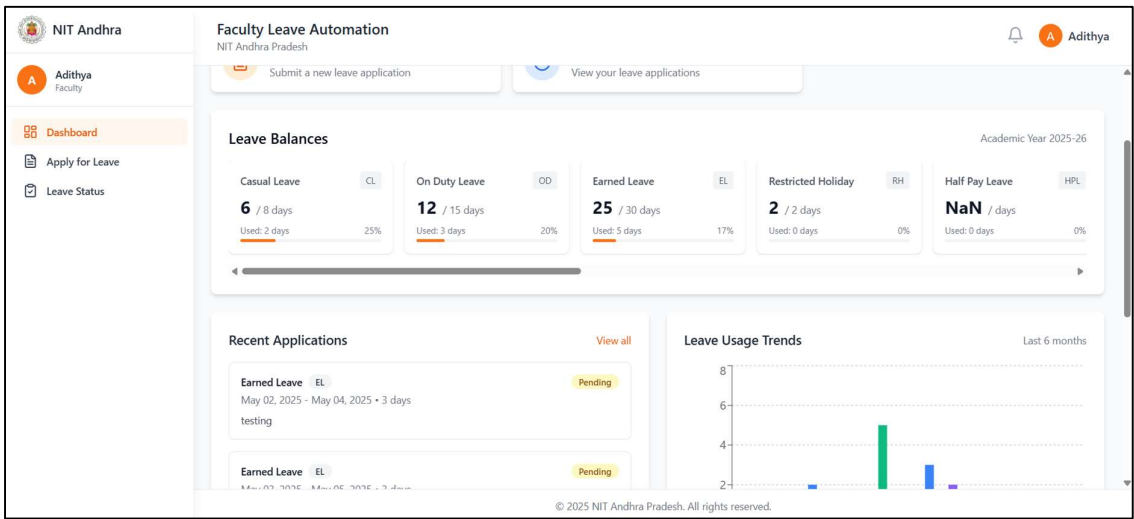


Figure 4.2.2.1 Leave balance and history

4.2.3 Leave Status Tracking

Faculty can track their leave status in real-time (e.g., Pending, Approved, Rejected). The process is shown in Figure 4.2.3.1 and Figure 4.2.3.2.

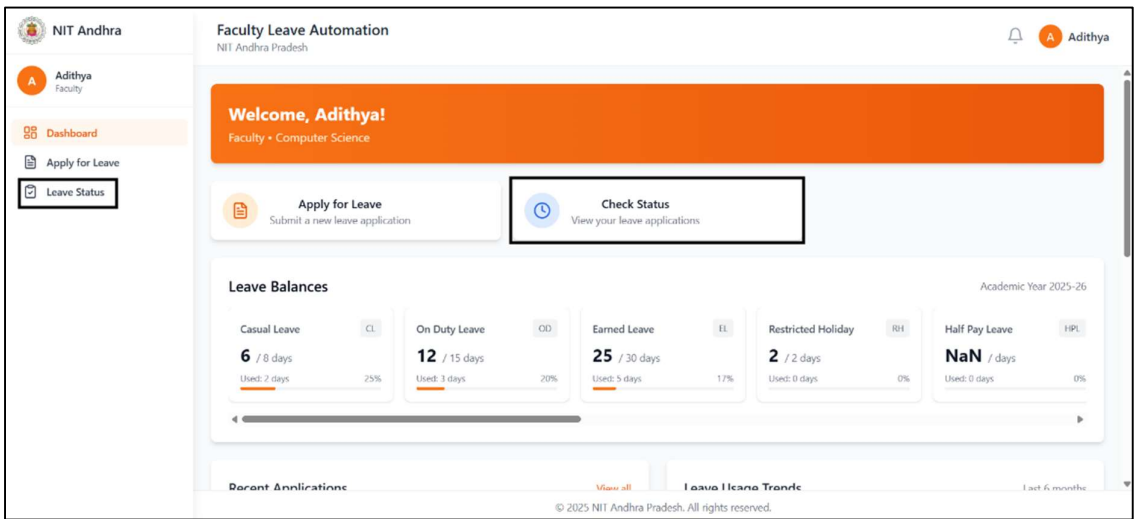


Figure 4.2.3.1 Check Status Navigation

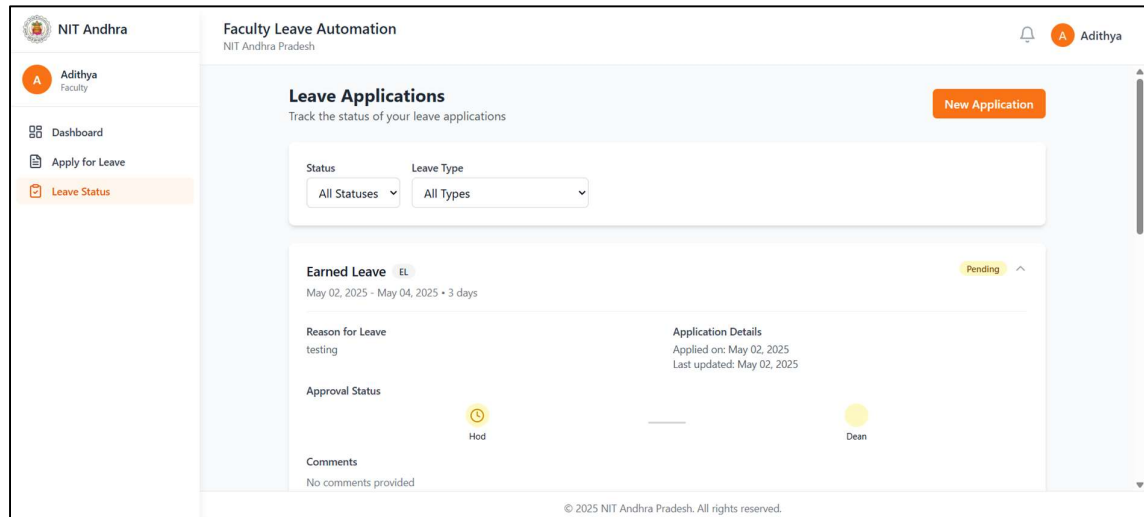


Figure 4.2.3.2 Leave Status Page

4.2.4 Notification Alerts

Faculty receive notifications at each stage of the leave application process. Alerts, as depicted in Figure 4.2.4.1, are visible to them.

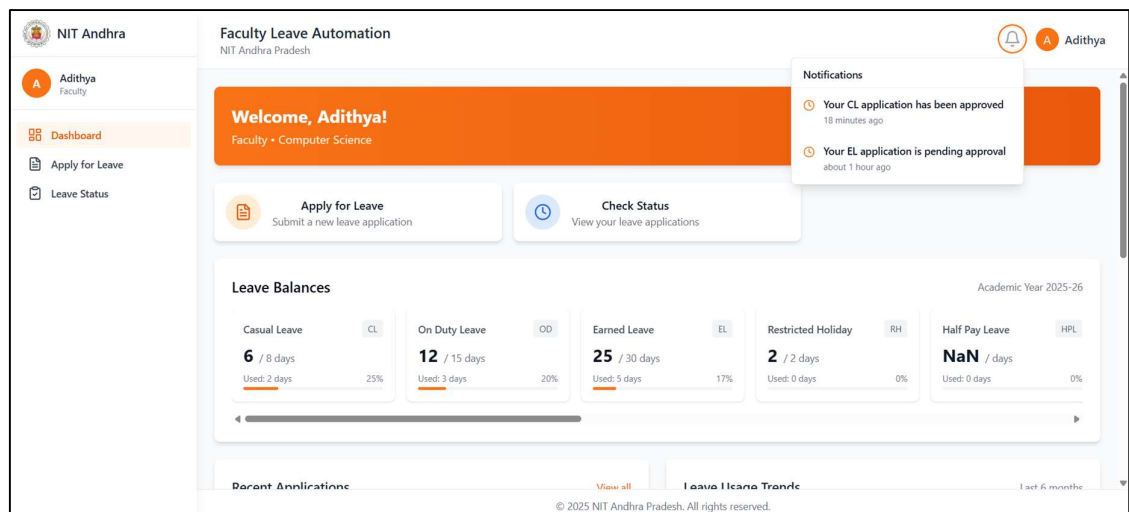


Figure 4.2.4.1 Notification Alerts Option

4.3 Approver side Features

4.3.1 All Faculty Features

This feature emphasizes that users with approver roles (e.g., Heads of Departments, Deans) retain all the functionalities of a regular faculty

member within the system. In other words, approvers can also:

- Submit their own leave applications using the customized forms.
- Track the status of their submitted leave applications.
- View their own leave balances and history.
- Receive notifications and alerts related to their own leave requests.

4.3.2 Dashboard View of Pending Requests

Approvers can view a dashboard of pending leave requests. The navigation to this dashboard is shown in Figure 4.3.2.1 and Figure 4.3.2.2.

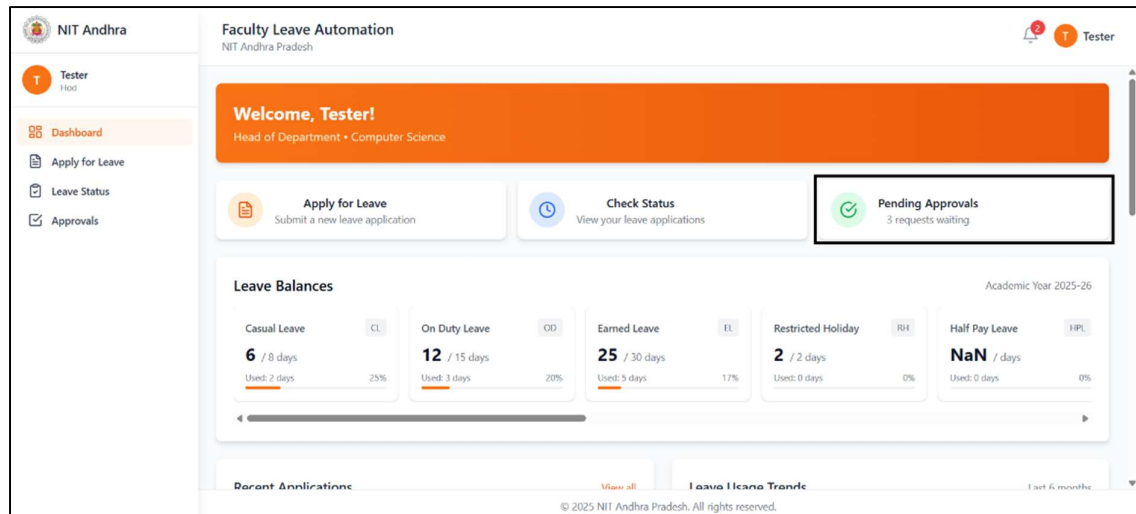


Figure 4.3.2.1 Pending Approvals Navigation

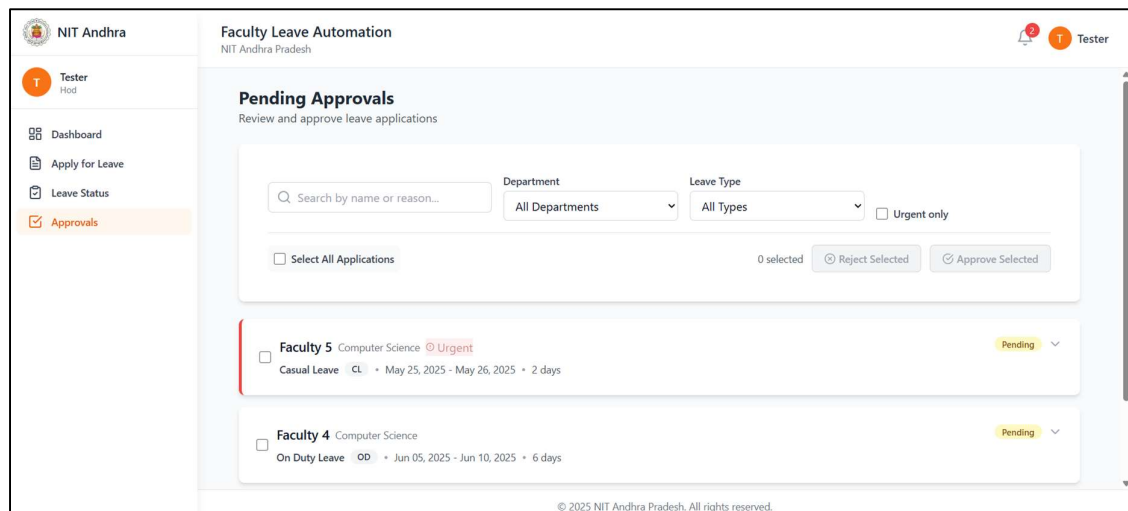


Figure 4.3.2.2 Pending Approvals Page

4.3.3 Approve / Reject Leave Requests

Approvers can approve or reject leave requests, as shown in Figure 4.3.3.1.

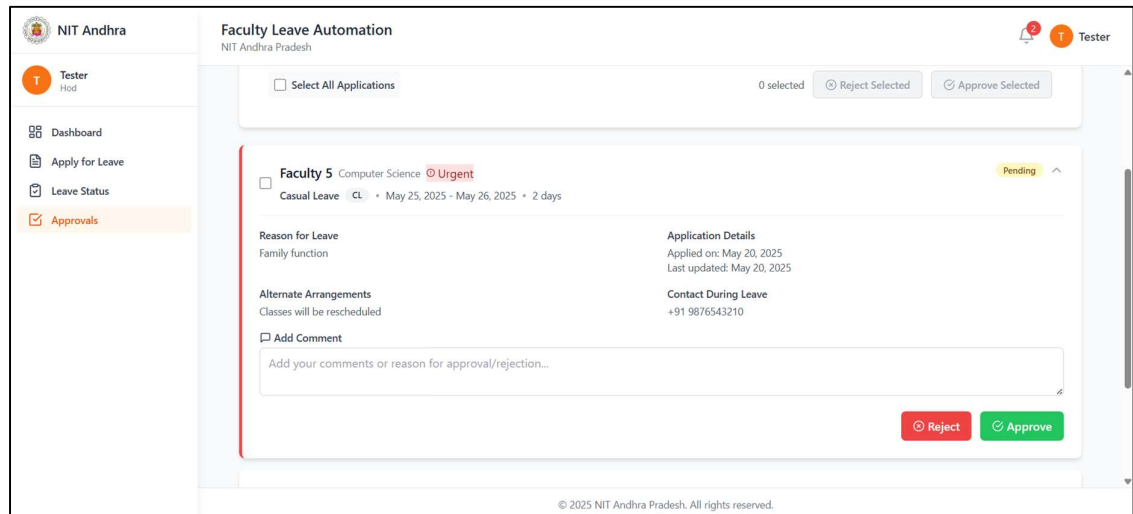


Figure 4.3.3.1 Reject / Approve Leave Request

4.3.4 Remarks and Comments on Requests

Approvers can add remarks and comments to leave requests, as shown in Figure 4.3.4.1.

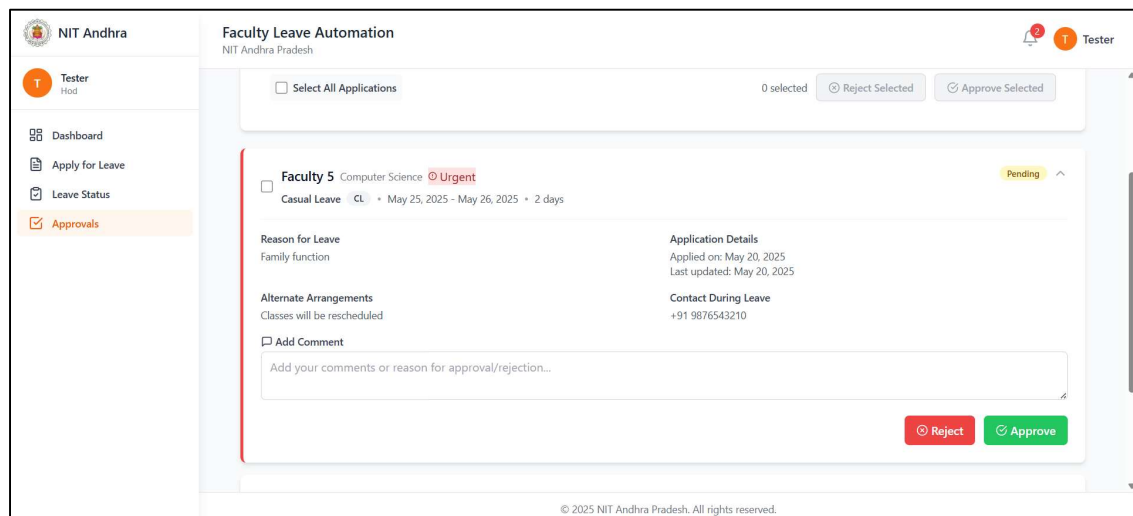


Figure 4.3.4.1 Add Comment on Leave Request

4.3.5 Notification for New Requests

Approvers receive notifications for new leave requests. Figures 4.3.5.1 and

4.3.5.2 illustrate how to check these new request notifications.

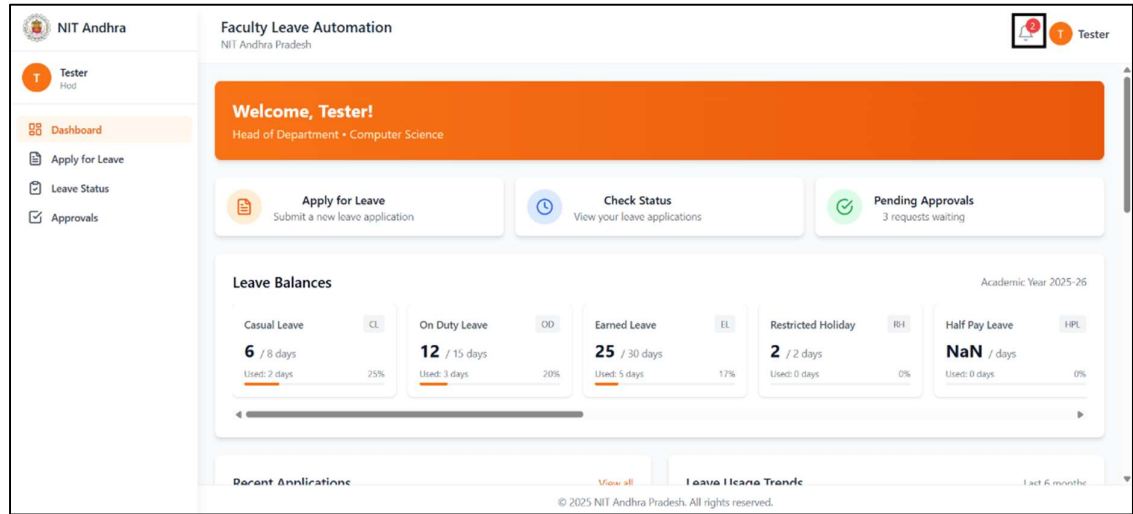


Figure 4.3.5.1 Notifications Alerts Option

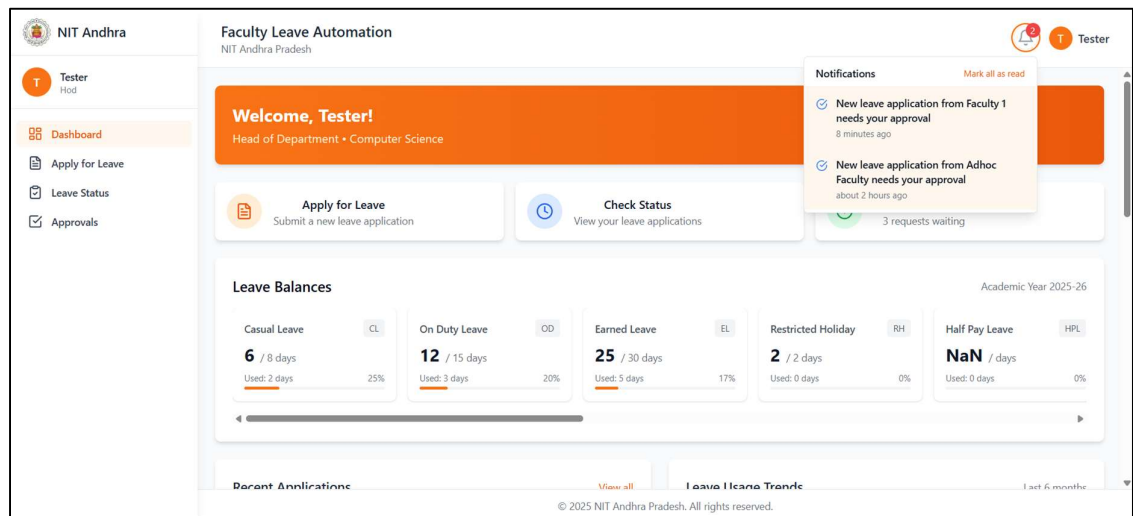


Figure 4.3.5.2 New Leave Requests Alerts

CHAPTER 5

5.1 Working Procedure

The system follows a structured workflow to ensure a seamless and transparent leave management process, as illustrated in the accompanying diagram. The primary steps involved are:

1. **Login:** Users (faculty and approvers) access the system through a secure login portal.
2. **OTP Verification:** Upon entering their institute email, users undergo a One-Time Password (OTP) verification process sent to their registered email address, enhancing security.
3. **Dashboard:** After logging in, users see their specific dashboard based on their role. Approvers see information like pending requests, and faculty see their leave balances.
4. **Submit Leave:** Faculty members initiate a leave request by navigating to "Apply for Leave." They will see a form that changes based on the type of leave and they provide all the necessary details according to NIT Andhra Pradesh's leave rules.
5. **Approval Flow:** Once a leave request is submitted, it goes through the defined approval steps, which usually follow a hierarchy. Approvals are needed from different levels depending on the leave type and the faculty member's department. Typically, it proceeds as follows:
 - Submission to the Head of Department (HoD).
 - Forwarding to the Dean (if needed).
 - Further routing to the Registrar and potentially the Dy. Director for specific leave types or faculty categories (e.g., Adhoc Faculty).
6. **Approval Status:** Both the faculty member who applied and the approvers can see the current status of the leave request at each stage. The system updates the status (e.g., Pending, Approved, Rejected) as it moves through the process.
7. **Notifications:** Users receive in-app notifications when important actions occur, such as upon submission of a request, when there are approval updates, and when a final decision is made.
8. **Log Entry:** The system records all actions related to a leave request, including submissions, approvals, rejections, and comments.
9. **Leave Balance Update:** When a particular leave request is approved, it automatically updates his/her corresponding leave balance.

CHAPTER 6

6.1 Programming Languages

6.1.1 TypeScript

- **Description:** TypeScript is a language that is a superset of JavaScript. It helps us write code that's easier to manage. It allows us to be more specific about the types of data we're using, which helps us to catch mistakes early [1].
- **Usage:** We're using TypeScript for both frontend and backend. This helps our code to keep organized and makes it easier to add more features later on without any errors.

6.2 Frontend Technologies

6.2.1 React.js

- **Description:** React.js is a JavaScript library used to build dynamic and interactive user interfaces. It allows us to create reusable components and updates the page efficiently [2].
- **Usage:** We are using React.js to build dynamic user interface for our project. This allows faculty and approvers to navigate the different options and features across project easily.

6.2.2 Vite.js

- **Description:** Vite.js is a modern JavaScript build tool that optimizes web development by providing built-in support for common web features, which makes starting and updating websites much faster and easier [3].
- **Usage:** We're using Vite.js with React.js part of our project to make things smoother. We can see changes instantly when we're coding and it helps us to create the final website easily.

6.2.3 Tailwind CSS

- **Description:** Tailwind CSS is a utility-first CSS framework that optimizes web development by providing a set of pre-designed utility classes. It allows easy customization of designs without writing additional CSS.
- **Usage:** We're using Tailwind CSS to style all the frontend components. This helps ensure a consistent visual style and speeds up the UI development through application of pre-designed utility classes.

6.2.4 HTML

- **Description:** HTML (Hypertext Markup Language) is the standard markup language for creating web pages. It is used to structure the content on the web. It organizes text, images, links and media using tags and elements that browsers can interpret.
- **Usage:** We're using HTML because it provides the basic structure for all pages, including forms for leave applications, tables for displaying leave data, and the navigation elements of the site.

6.3 Backend Technologies

6.3.1 Node.js

- **Description:** Node.js is a runtime environment that allows JavaScript to run outside the browser, enabling backend development [4]. It is a single-threaded, asynchronous, and also non-blocking.
- **Usage:** We're using Node.js to build the server-side logic for our web application, which handles API requests from frontend, manages application data, and interacts with the database to store and retrieve information related to leave applications and user data.

6.3.2 Express.js

- **Description:** Express.js is a flexible Node.js web application framework that provides a list of features for building web applications. It simplifies the server-side applications development by providing an easy-to-use

API for routing, middleware, and HTTP utilities [5].

- **Usage:** We're using Express.js to create the RESTful APIs for our project by defining the specific endpoints for functionalities like user authentication, leave applications submission, approval process, and data retrieval.

6.4 Database

6.4.1 PostgreSQL

- **Description:** PostgreSQL is an open source object-relational database system used to store and manage information in an organized manner. It is well-known for its reliability and data integrity [6]. It's ACID-compliance guarantees keeping our data consistent. It also works on all major operating systems.
- **Usage:** We're using PostgreSQL to securely and consistently store all the application's data such as user details, leave applications, leave balances, notifications, department information, and approval workflow. This ensures our data is properly saved and remains consistent.

CHAPTER 7

7.1 Conclusion

Our "**Faculty Leave Automation**" project has created a digital solution to manage leave approval process at NIT Andhra Pradesh, which is much better than the old paper system. It makes things automatic, clear, and easy for faculty and approvers. This will lead to faster processing, fewer delays, and a better experience for everyone, helping the institution be organized and more efficient.

The system makes leave management easier and faster than existing process. It lets everyone track their leave requests, makes approvals way more clear and easier. By having different access for different roles and helping with urgent leave requests, it simplifies the leave approval process and makes decisions easier. Automating the present manual work reduces the burden on staff and provides useful information for future improvements.

7.2 Further Improvements

The following mentioned improvements can be made in the future to optimize functionality way better:

- **Branch-wise Reports:** Implement branch-wise leave reports to provide Deans and HoDs with insights into leave patterns within their departments.
- **Academic Calendar Integration:** Integrate with the institute's academic calendar to provide a detailed view of faculty availability which makes planning easier and helps avoid problems.
- **Enhanced Delegation of Authority:** Expand delegation features to allow approvers to transfer specific approval responsibilities, ensuring workflow continuity.
- **Admin Portal:** Develop a dedicated admin portal for streamlined user and system configuration, as well as data management.
- **Enhanced Analytics:** Expand the analytics dashboard with customizable reports and visualizations to provide deeper insights into leave trends.

7.3 References

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